

Harshit Singhai

SOFTWARE ENGINEER

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Work Experience

Deloitte Consulting USI

Gurugram, India

AI ENGINEER CONSULTANT - AI & DATA

Feb. 2022 - Ongoing

- Architected end-to-end **GenAI & cloud ML microservices** on **AWS** for Fortune 500 clients, ensuring high availability and scale
- Developed production **Agentic RAG chatbots** using **Python** and **AWS Bedrock**, integrating tool execution and multi-hop reasoning
- Built **Agentic RAG** with **proactive callbacks, order tracking, and order management**, reducing handoffs to human agent by **35%**
- Improved retrieval quality using **query decomposition** and **HyDE**, increasing relevant document recall for multi-turn queries
- Built data ingestion on **AWS ECS** to chunk markdown articles into vector embeddings and enable **hybrid search (semantic + BM25)**
- Validated LLM accuracy using **LLM-as-a-Judge** automated frameworks to compute hallucination, groundedness, and answer relevancy metrics
- Integrated **LangSmith** for full-stack LLM observability, tracing multi-step agent execution and monitoring real-time token latencies
- Engineered serverless **GenAI agents** on **AWS** to convert Jira user stories into draft PRs, **cutting feature delivery timeline by 40%**
- Built **Agentic RAG** with **Python, LangGraph & Bedrock** to replace Dialogflow, enhancing multi-turn intent handling and response accuracy
- Engineered high-throughput backend infrastructure with **FastAPI & PostgreSQL**, optimizing API latency by **40%** for scaled workloads
- Integrated **Microsoft Presidio** for automated PII detection and masking, ensuring strict data privacy and compliance in LLM pipelines
- Implemented **Bedrock Guardrails & LLM Guardrails** to enforce context-aware safety, block prompt injections, and filter harmful content
- Deployed **semantic caching** using vector similarity indexing to eliminate redundant LLM API requests and accelerate response latency

Quantive

Indore, India

SOFTWARE ENGINEER - PLATFORM

Aug. 2021 - January 2022

- Optimized microservices leading to **30% reduction** in memory usage & processing time through memory profiling & improved code readability
- Enhanced Pandas pipelines with vectorized operations and memory-efficient data types reducing processing time by **10%** for large datasets
- Increased data throughput by **40%** using Python Celery **producer-consumer architecture** for async processing and ClickHouse for storage

Deepsource

Bengaluru, India

SOFTWARE ENGINEER - MACHINE LEARNING

December 2020 - June 2021

- Developed a recommendation system using XGBoost regression to rank code issues by likelihood of being resolved based on historical patterns
- Created 20k+ personalized ML models, one for each repository, to predict high-impact code issues, saving developers 10 hours/week
- Built adaptive ML retraining pipelines to prevent model decay, updating ML model only when new data improved precision and relevance

Programming Skills

Server Side	Python, JavaScript (React, Node.js), FastAPI, REST API
Data Engineering	PySpark, Pandas, AWS Glue, SQS, S3, Athena, SQL
Machine Learning	Strands SDK, Qdrant, Bedrock Knowledge Bases, AgentCore, Langfuse, MCP, Multi-Agent, Bedrock, Scikit-Learn, Keras, MLOps
Database	SQL/NoSQL, PostgreSQL, MongoDB, AWS DynamoDB, Redis
Other	AWS, Docker, Git, Problem Solving, Data Structures Algorithms

Projects

- <https://nemo-ai.netlify.app/>: **NemoAI** - Cut Your Feature Delivery Time in Half. Nemo AI converts your Jira Stories into ready-to-review PR
- fictionally-irrelevant.vercel.app/posts/side-projects-2023: Shipped and deployed **20+** user-facing side projects, reaching over **1k users** globally
- <https://fictionally-irrelevant.vercel.app> : **Authored 72+** in-depth technical blog posts, delving into diverse areas like **AI, Cloud, Data, and web3**
- **MLH Fellowship**: Selected among the top **144** out of 30,000 applicants; contributed to **BentoML** open-source Python project

Accomplishments & Awards

2023	Cohort Lead , Deloitte AI Academy Explorer Program: Guided 80+ practitioners on Datacamp ML track	Deloitte USI
2022	Applause Award , recognized by Senior Leadership for delivering outstanding client outcomes	Deloitte USI
2019	Mentored , sophomore students for Python: Conducted and managed 30+ labs sessions and tutorials	Bennett University
2019	National Finalist , Smart India Hackathon: Enhanced Kotak Mahindra Bank's virtual assistant Keya	Bhopal

Education

Bennett University

Delhi NCR, India

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING (CSE) - 7.98 GPA

Aug. 2016 - Aug. 2020

- Main coursework: **Data Structures and Algorithms, Artificial Intelligence, Machine Learning, Cloud Computing, Big Data**